DANIEL CHEN

Austin, TX · (832) 643-8245 · dyc394@my.utexas.edu · linkedin.com/in/danielchen05

EDUCATION

The University of Texas at Austin — Austin, TX
B.S Electrical and Computer Engineering — Expected May 2028

- **Overall GPA:** 3.62/4.00
- **Relevant Coursework:** Electromagnetic Engineering, Circuit Theory, Embedded Systems
- **Honors:** University of Texas Honors Recognition — Fall 2024, Spring 2025, Spring 2026

SKILLS

Hardware: Oscilloscopes, Multimeters, Soldering, PCB Design, GPIO, Optical Systems, Electrical Characterization
Software: MATLAB, Python, C, C++, ARM Assembly, LabVIEW, LTspice, KiCad, Onshape

EXPERIENCE

Undergraduate Research Assistant - *Nanomanufacturing & Nanolithography Lab* — April 2026 - Present

- Configured and integrated two laser-based photolithography and holographic fabrication systems by incorporating optical components and experimental hardware for volumetric printing research.
- Developed and tested Python interface for optical shutters, achieving >99% reduction in timing error and enabling microsecond-level accuracy compared with manual operation.
- Characterized 10+ conductive printed features through electrical testing and materials evaluation to support development of functional structures for advanced manufacturing applications.
- Designed custom 3D-printable CAD housings for two board-level imaging systems, incorporating PCB/component clearances, optical-post mounting, threaded hardware, and thermal considerations.
- Optimized a fluorescent dye-resin alignment solution to visualize laser projections during optical alignment.

ACADEMIC PROJECTS

Dungeon Crawler RPG (C++) - *Software Design and Implementation* — October 2025 - November 2025

- Architected a modular C++ text-based RPG using inheritance, virtual functions, and runtime polymorphism.
- Implemented combat, inventory, and persistence systems using STL containers with RAII, reducing errors by 25%.
- Developed a save/load system using file I/O that preserves user progress, supporting 20+ simultaneous objects.

Pong Game - *Embedded Systems* — January 2025 - May 2025

- Constructed a bare-metal C game engine on the MSPM0G3507 microcontroller, demonstrated to 100+ peers.
- Implemented collision detection for dual-ball gameplay, with a slide potentiometer for responsive user input.
- Optimized for an 80 MHz microcontroller, achieving smooth animations under hardware constraints.
- Added a multilingual UI supporting multiple languages, enhancing accessibility without an operating system.

Bike Optimization and Reporting Consultant - *Client: CapMetro* — January 2025 - May 2025

- Proposed new station locations improving service for 50,000+ UT Austin students based on convenient utilization.
- Worked in a team, analyzing CapMetro's Bikeshare, identifying high-demand areas around the city of Austin.
- Translated 10,000+ entries of bike station data using Excel into recommendations for non-technical stakeholders.

ACTIVITIES

Student Engineering Council - *Corporate Chair* — May 2026 - Present

- Lead corporate sponsorship outreach efforts for UT's SEC, managing relationships with 250+ industry contacts.
- Secured $22,000+ in corporate sponsorship funding through targeted outreach and partnership development efforts.
- Created sponsorship materials and outreach campaigns used to support fundraising and corporate relations.

Management Information Systems Association - *Logistics Director* — January 2025 - Present

- Manage back-end member data ensuring active participation and contributions to the organization.
- Support executive operations by optimizing internal systems and processes that enhance event efficiency.
- Oversaw data-driven strategies to analyze member engagement, resulting in a 20% increase in event attendance.